

Methodology & Results — Redrob Ranking Engine

Slide 1: Problem Statement

Traditional ATS ranking is broken.

- Boolean keyword matching produces false negatives at scale
- A candidate who "built highly scalable applications with React at Meta" ranks below someone who stuffed "Senior Frontend Engineer" five times
- Keyword filters cannot: see career trajectory, tell a top performer from a mediocre applicant, or explain *why* anyone was shortlisted

Goal: Replace keyword filters with a hybrid retrieve → score → explain pipeline that is reproducible, explainable, and LLM-efficient.

Slide 2: System Architecture

Offline (run once per dataset, ~41 min):

- 100,000 candidate profiles → Candidate Document Builder → `all-MiniLM-L6-v2` embeddings (384-dim) → FAISS IndexFlatIP → `candidate_vectors.npy`
- Structured Feature Extractor → trajectory/stability/platform/cert sub-features → `candidate_features.parquet`

Online (run once per JD, target < 60s):

1. JD Parser (1 LLM call) → `structured_jd_intent`
2. JD Embedder (same model as index)
3. ANN Retrieval → FAISS top-500
4. Multi-Signal Scoring (deterministic, 1.2s for 500 candidates)
5. Ranker → top 20 by `composite_score`
6. Explanation Generator (batched LLM calls)
7. Output → `ranked_output.csv` / `.json`

Key design principle: LLM used exactly twice per JD — understanding the JD once, and narrating 20 already-computed decisions. Pure math everywhere it has to run at scale.

Slide 3: Scoring Methodology

Five independent, deterministic sub-scores:

Sub-score	Weight	What it measures
B1 — Semantic Skill Match	0.35	Embedding similarity + JD skill coverage with semantic equivalence
B2 — Trajectory & Seniority Fit	0.25	Seniority alignment (penalized both ways) + promotion history + stretch readiness
B3 — Career Stability	0.15	Average tenure, job-hopping pattern, education-tier tiebreaker
B4 — Platform Activity & Intent	0.20	Active job-seeking signals, recruiter demand, hire reliability
B5 — Certification Bonus	+0.05 (cap 0.10)	Relevance- and recency-weighted certification boost

Composite: $B1 \times 0.35 + B2 \times 0.25 + B3 \times 0.15 + B4 \times 0.20 + B5 \times 0.05$

Every sub-score is persisted — nothing is collapsed before it's shown.

Slide 4: Indexing Performance

Metric	Value
Dataset size	100,000 candidates
Total indexing time	41 minutes 31 seconds
Embedding model	<code>all-MiniLM-L6-v2</code> (384-dim)
Batch size	256
Threading	3-level: work-stealing pool + per-core workers + embedding pool
Records loaded	100,000 (100%)
Records skipped	0
Index artifacts	<code>candidate_vectors.npy</code> , <code>faiss_index.bin</code> , <code>candidate_features.parquet</code> , <code>candidate_id_map.json</code>
Total disk size	~300 MB

Scalability note: Indexing is O(n) in the number of candidates. Doubling to 200K would take ~83 min on the same hardware.

Slide 5: Retrieval & Query Performance

Single JD query pipeline (Senior Data Engineer):

Stage	Time	Notes
JD parsing	~1.5-2s	1 LLM call via Groq (llama-3.3-70b-versatile)
ANN retrieval	~0.1s	FAISS top-500 from 100K
Multi-signal scoring	~1.2s	500 candidates × 5 dimensions
Explanation generation	~60-80s	Batched LLM calls for top 20 (with TPD retries)
Total	~80-140s	Varies with Groq rate-limit retries

Optimization highlight: Scoring was 116s before skill-vector caching. Pre-encoding JD skill vectors once and caching per-candidate skill vectors reduced this to **1.2s** — a **100× speedup**.

Slide 6: Quantitative Evaluation (All 5 Golden JDs)

Metric	DE	HR	Analyst	PM	Accountant
Grounding	20/20	20/20	20/20	16/20	20/20**
Strong matches	9	13	20	13	—
Moderate matches	11	7	0	3	—
Fallback	0	0	0	4	0
Contamin.@100	0	0	0	0	0
Precision@20	15/15	15/15	15/15	15/15	15/15
Semantic range	[0.657, 0.668]	[0.663, 0.676]	[0.692, 0.698]	[0.677, 0.689]	[0.679, 0.689]
Composite range	[0.668, 0.721]	[0.671, 0.734]	[0.701, 0.741]	[0.670, 0.737]	[0.636, 0.725]

*HR/Analyst had 0/20 in prior TPD-limited run. With GROQ key + fallback fix: 20/20, 20/20.
**Accountant re-run with llama-3.1-8b-instant (batch_size=1) after TPD reset — 20/20 grounded via model-switch workaround.

Grounding summary: 96/100 LLM-grounded, 4/100 structured fallback (PM only, TPD-limited mid-run), 0 nulls, 0 silent failures across all 5 JDs and 100 candidates.

Key: 0 contamination across all JDs. Domain purity is near-perfect (DE, HR, Analyst are 100% on-domain; PM includes relevant Ops Manager titles; Accountant has mild title noise). NDCG improved with stronger bge-small-en-v1.5 model.

Slide 7: Ablation & Keyword-Stuffer Detection

Ablation study — removing one dimension at a time:

Config	Overlap with baseline Jaccard	
Baseline (all 5 dims)	20/20	—
No B1 (semantic)	19/20	0.905
No B2 (trajectory)	9/20	0.290
No B3 (stability)	15/20	0.600
No B4 (platform)	14/20	0.538
No B5 (cert)	20/20	1.000

Finding: Trajectory (B2) remains the single most impactful dimension — removing it still causes 55% of the top-20 to change.

Keyword-Stuffer Case Study:

- Candidate: CAND_0066999 (low trajectory but strong title match)
- **With B2 trajectory scoring:** rank 501 (not in top 500 retrieved)
- **Without B2:** rank 2 (jumps 499 positions into rank 2)
- **B2 score:** 0.591 (low = weak career progression despite matching title)
- **Interpretation:** Title-level match but weak career progression. When trajectory scoring is removed, this candidate surges 499 positions into the top 2. Exactly the signal the system is designed to catch.

Slide 8: Limitations & Future Work

Current limitations:

- **No labeled ground truth** — validation relies on manually-constructed golden JDs and anchors
- **CPU-sized embedding model** — BAAI/bge-small-en-v1.5 (384-dim) still has narrow semantic spread (~0.010); bge-base-en-v1.5 (768-dim) would improve differentiation
- **Synthetic dataset artifacts** — uniform title/skill distribution constrains certain scoring decisions
- **Anchor recall shifted** after re-index with new model — some anchor candidates no longer in top 100; expected with model change
- **Groq TPD limited** — PM (4/20 fallback) hit Free Tier TPD mid-run; Accountant required model-switch workaround (llama-3.1-8b-instant with batch_size=1 to respect TPM limits). Structured fallback protects all remaining cases
- **No resume parsing** — consumes pre-structured JSON only

With more time:

- Run full 5-JD × 5-ablation grid for each JD
- Switch to BAAI/bge-base-en-v1.5 (768-dim) for better role differentiation (requires re-indexing)
- Add BM25 hybrid retrieval to harden recall on edge-case JDs
- Add LLM-as-judge secondary reference ranking for independent rank correlation
- Integrate feedback loop from recruiter actions into weight tuning
- Add raw PDF resume ingestion
- Upgrade GROQ API key to Dev Tier to avoid TPD limits across multi-JD evaluation runs

"BAAI/bge-small-en-v1.5 (384-dim) was chosen over all-MiniLM-L6-v2 for its stronger asymmetric query-to-document retrieval at 100K scale, documented as the PRD Risk 1 fallback path. Domain separation was validated: DE/HR/Analyst show 100% on-domain titles in top 20."